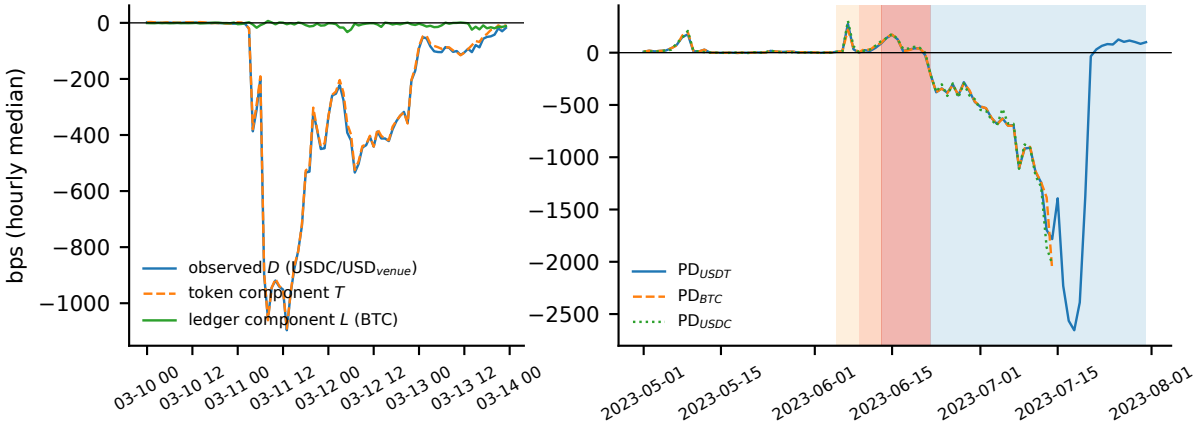
(a) Token failure: USDC, March 2023
bps (hourly median)
0
−200
−400
−600
−800
−1000
observed D (USDC/USD$_{venue}$)
token component T
ledger component L (BTC)
03-10 00
03-10 12
03-11 00
03-11 12
03-12 00
03-12 12
03-13 00
03-13 12
03-14 00
(b) Ledger failure: Binance.US dollar, 2023 (shaded: rail events)
0
−500
−1000
−1500
−2000
−2500
PD$_{USDT}$
PD$_{BTC}$
PD$_{USDC}$
2023-05-01
2023-05-15
2023-06-01
2023-06-15
2023-07-01
2023-07-15
2023-08-01